

Data C8: Foundations of Data Science

UC Berkeley, Summer 2025

First things first

A message from DSUS

In support of [our commitment to making Data Science education inviting, engaging, and respectful for people of diverse identities, backgrounds, experiences, and perspectives](#), I am relaying this message from Data Science Undergraduate Studies (DSUS):

Device Lending options

Students can access device lending options through the Student Technology Equity Program STEP program.

Data Science Student Climate

[Data Science Undergraduate Studies](#) faculty and staff are committed to creating a community where every person feels [respected, included, and supported](#). We recognize that incidents may happen, sometimes unintentionally, that run counter to this goal. There are many things we can do to try to improve the climate for students, but we need to understand where the challenges lie. If you experience a remark, or disrespectful treatment, or if you feel you are being ignored, excluded or marginalized in a course or program-related activity, please speak up. Consider talking to your instructor, but you are also welcome to contact Executive Director Christina Teller at cpteller@berkeley.edu or report an incident anonymously through this [online form](#).

Community Standards (Ed)

Ed is a formal, academic space. Posts in this forum must relate to the course and be in alignment with [Berkeley's Principles of Community](#) and the [Berkeley Campus Code of Student Conduct](#). We expect all posts to demonstrate appropriate respect, consideration, and compassion for others. Please be friendly and thoughtful; our community draws from a wide

spectrum of valuable experiences. Posts that violate these standards will be removed.

Course climate (a message from me)

Learning is cooperative

We encourage you to discuss course content with your friends and classmates while working on your assignments. No matter your academic background, you will learn more if you work alongside others than if you work alone. Ask questions, answer questions, and share ideas liberally. If an emergency takes you away from the course for an extended period, or if you decide to drop the course for any reason, let us know immediately. This includes informing the instructor, your lab TA and your project partner.

Some words of encouragement and perspective

Make a strong effort on every assignment! If you get stuck, read the textbook and go over the lectures and lab discussion.

When you need help, reach out to the course staff using Ed, in office hours, and/or during labs. Here, you can discuss any doubts that you have. You are not alone in Data 8! Instructors and staff are here to help you succeed. We expect that you will work with integrity and respect for other members of the class, just as the course staff will work with integrity and respect for you.

Finally, know that it's normal to struggle. Berkeley has high standards, which is one of the reasons its degrees are valued. Everyone struggles, even though many try not to show it. Even if you don't fully master everything covered, you can build on what you did learn, and oftentimes, things will just naturally click into place over time. You aren't expected to be perfect! **It's OK not to get an A!** I certainly didn't get A's in everything.

We are here for you

It can be very tough to be a student at this school! There are applications to clubs and grade requirements to declare majors, which are two things I did not have to experience as an undergrad. Some of you are navigating other challenges, like being a parent or commuting long distances from home to campus. I've learned these things from my students during my time so far here teaching, and if you have any other things you'd like to share with me about your experiences or if you just need someone to talk to about your academic struggles or your future path, I can be there for you. The tutors and TAs might be an even better resource than myself for some topics, because they are students just like you. So feel free to have conversations with them as well. They can also tell you what being an Academic Student Employee (ASE) is like.

With regards to reports of sexual misconduct/violence/assault, you may speak with me (and with the course staff, for that matter), but know that we will need to report the discussion to the Title IX officer. This is detailed below.

As UC employees, I and the ASEs are “Responsible Employees” and are therefore required to report incidents of sexual violence, sexual harassment, or other conduct prohibited by University policy to the Title IX officer. We cannot keep reports of sexual harassment or sexual violence confidential, but the Title IX officer will consider requests for confidentiality. Note that there are confidential resources available to you through UCB’s PATH to Care Center, which serves survivors of sexual violence and sexual harassment; call their 24/7 Care Line at 510-643-2005.

About the course

Course description

Foundations of Data Science combines three perspectives: inferential thinking, computational thinking, and real-world relevance. Given data arising from some real-world phenomenon, how does one analyze that data so as to understand that phenomenon? The course teaches critical concepts and skills in computer programming and statistical inference, in conjunction with hands-on analysis of real-world datasets, including economic data, document collections, geographical data, and social networks. It also delves into social issues surrounding data analysis, such as privacy and study design.

Prerequisites

The curriculum and format are designed specifically for students who have not previously taken statistics or computer science courses. Students with some prior experience in either statistics or computing are welcome to enroll and often find that this course offers a new perspective that blends computational and inferential thinking. Students who have taken several statistics or computer science courses should instead take a more advanced course like [Data 100](#).

Materials & resources

Our primary text is an online book called [Computational and Inferential Thinking: The Foundations of Data Science](#). This text was written for the course by its founding instructors: Prof. Ani Adhikari, Prof. John DeNero and Prof. David Wagner.

The computing platform for the course is hosted at data8.datahub.berkeley.edu. Students find it

convenient to use their own computer for the course. If you do not have adequate access to a personal computer, you can reach out to the Student Technology Equity Program STEP program.

Support in learning the content

You are not alone in this course; the staff and instructor are here to support you as you learn the material. It's expected that some aspects of the course will take time to master, and the best way to master challenging material is to ask questions.

Office hours

Visit the tab on this website (left sidebar) to see our weekly office hour schedule. Both the instructor and the staff will be hosting weekly office hours.

Ed discussion forum

For questions about course material outside of Office Hours, students will use our course discussion forum, Ed Discussion ("Ed"). Ed is a great resource for asking questions about course material and getting help from the professor, course staff, and other students. Ed is a formal, academic space. Posts in this forum must relate to the course and be in alignment with [Berkeley's Principles of Community](#) and the [Berkeley Campus Code of Student Conduct](#). We expect all posts to demonstrate appropriate respect, consideration, and compassion for others. Please be friendly and thoughtful; our community draws from a wide spectrum of valuable experiences. Posts that violate these standards will be removed.

Group tutoring

Starting in the second week of the semester, small-group tutoring sessions will be available for students wanting additional support. These sessions are meant primarily to develop confidence with concepts that have been taught in previous weeks. In these sessions, a tutor will guide students in groups of no more than five through a specialized tutor worksheet which covers past concepts. Details about sign-ups will be available during the first week of the semester.

Course components

Lecture

Lectures will be held daily in Dwinelle 155. On Mondays, Tuesdays, Wednesdays and

Thursdays, they start at 11:00am and end at 12:00pm; on Friday they start at 11:00am and may run until 1:00pm. Students are expected to attend, and if you are unable to make it on a given day or missed something during the lecture, recordings will be made available on bCourses Media Gallery. In addition, for each lecture, the slides and accompanying coding examples (called demos) will be provided on the course website.

Labs

Biweekly labs include a discussion worksheet covering recent material and a programming-based lab assignment that develops skills with computational and inferential concepts. The problems in lab are good preparation for similar (and harder) problems you will see on your homework assignments and exams! Lab assignments will be released on Mondays and Wednesdays each week and are due that same day.

More on lab

Labs are two hours long and take place on Mondays and Wednesdays every week, following the lecture for that day.

- The first hour focuses on a discussion worksheet and group problem-solving. We encourage no use of technology during this portion of the lab (all materials required will be available at the lab section).
- The second hour is dedicated to completing a programming-based assignment.
- 80% of lab credit will be attendance-based.
- The remaining 20% of credit will be awarded for submitting the programming-based assignment to Gradescope by the deadline (later in the day) with all test cases passing. Discussion worksheets do not need to be submitted.
- Active participation in the discussion and lab is required to earn attendance credit.
- Lab sessions will not be webcast or recorded.

How Should You Submit The Programming-Based Assignment?

[Here](#) is a video for how to submit the programming assignment to Gradescope. Before posting about code errors, please check our [debugging page](#)! We have listed common errors and reasons why they might be coming up for you.

Asking for Help

You can ask us about lab assignments in office hours or on Ed in the dedicated lab thread for a given week! Remember to never post your code publicly; please make a private post if you have to post any code. *We also encourage you to review the course policies concerning collaboration*

and academic honesty.

Homework

Biweekly homework assignments are a required part of the course. You must complete and submit your homework independently, but you can discuss problems with other students and course staff. See the [Learning Cooperatively](#) section below. Homeworks will be released on Tuesdays and Fridays and due Friday and the Tuesday of the following week, respectively.

Remember, there are resources to help you all assignments, including your homeworks! You can ask questions on Ed in the dedicated homework thread (make sure to navigate to the subquestion thread). Remember to never post your code publicly; please make a private post if you have to post any code. *We also encourage you to review the course policies concerning collaboration and academic honesty.* You can also get help at office hours. We highly recommend getting started on the homework earlier, so that if you need help, you can attend OH before the deadline on Wednesday.

How to Submit Your Homework Assignment

[Here](#) is a link to a video on the assignment submission process (make sure you're logged into your Berkeley e-mail). Unlike labs, there are separate tests we run in addition to the ones you see in your Notebook! In order to pass these, make sure your code solves the general problem at hand and not only a specific case of the problem. As a reminder, it is your responsibility to make sure the autograder tests results in the notebook match the autograder results on Gradescope after you submit. Check out our [DataHub Guide](#) if you run into any issues working on Jupyter Notebook before posting on Ed.

Projects

Data science is about analyzing real-world data sets, and so you will also complete three projects involving real data. The experience of solving the problems in this project will prepare you for exams (and life in a data scientist role). On each project, you may work with a single partner; your partner must be from the lab you enrolled in. **Both partners will receive the same score and are equally responsible for the work submitted!!** You may also work on your own.

There are two projects throughout the semester. For each project, you will submit in two parts: first, a checkpoint, and then, the completed project. Don't share your code with anybody but your partner. You are welcome to discuss questions with other students but don't share the answers. If someone (who is not your partner) asks you for the answer, you might demonstrate how you would solve a similar problem.

The projects can seem long and difficult, but you are not alone! Come to office hours, post on Ed, and talk to your classmates. If you want to ask about the details of your solution to a problem, make a private Ed post and the staff will respond. If you're ever feeling overwhelmed or don't know how to make progress, email your TA or tutor for help. You can find contact information for the staff on the course website.

We *highly recommend* starting early, so on the day that the project is released, and doing each bit that you can with the material that you have been exposed to in lecture/lab up until that point. Continue in this fashion, working day-by-day, and you're more likely to have a stress-free experience!

Submitting your Project with a Partner

You may reference [this walkthrough video](#) on how to add partners on Gradescope! Make sure only one of you submits the project.

Exams

The midterm exam will be held on campus on Friday, July 18. Time and location TBD.

The final exam will be held on campus on Thursday, August 14. Time and location TBD.

More details will be released as the semester continues.

Grades

Grades will be assigned using the following weighted components. Every assignment is weighted equally in its category. For example, there are 2 projects, so each project is worth $(15 / 2)\% = 7.5\%$ of your grade.

Activity	Grade
Final	30%
Midterm	25%
Labs	25%
Projects	10%
Homeworks	10%

The instructor nor the TA will respond to any questions regarding grade bins or letter grades. Please consult Berkeleytime for historical distributions of grade bins!

Grades for Homeworks, Projects, and Labs will be posted on Gradescope within a timely manner after the assignment's due date. Solutions to the assignment and common mistakes will also be posted on Ed. It is up to you to check the solutions and request a regrade request before the **regrade deadline of 1 day after grade release**. Regrade requests can be made on Gradescope. Any regrade request past the deadline will not be looked at; this is to enforce the same deadline across all students, so again: please do not delay reviewing your work.

Submitting assignments

- All assignments (homework, labs, and projects) will be submitted on Gradescope.
 - Please refer to [this tutorial](#) for submitting assignments.
- The deadline for all assignments in this course is **8 PM PST** on the due date.
 - **Submissions after this time** will be **accepted for 24 hours and will incur a 30% penalty**.
 - Any submissions later than 24 hours after the deadline will not be accepted.
- If you submit an assignment within a day before the deadline, you will receive 5 bonus points on that assignment!

We understand that the submission process is new for many students taking the course. To account for this, we will do our best to accommodate submission-related issues (submitting to the wrong assignment, not saving files correctly, autograder timing out) through the second week of the course. After the second week, it is **your responsibility** to confirm you have submitted your work correctly, and we may penalize you for not doing so.

Drops

- **Homeworks:** Your two lowest homework scores will be dropped in the calculation of your overall grade.
- **Labs:** Your two lowest lab scores will be dropped in the calculation of your overall grade.
 - To use a lab drop, you must message your GSI prior to the start of your lab, informing them that you will not be in attendance. The earlier in advance, the more likely your drop will be approved.

If you have an ongoing situation that prevents you from completing course content, please contact the course instructor.

Assignment extensions

If you need to request an extension on an assignment, please fill out [this form](#). Submissions to the form will be visible only to staff! Please read the entirety of the form and its instructions carefully.

Extension requests need to be submitted at least 24 hours before the deadline to be considered. Some extension requests are subject to more detailed review and may require a meeting with course staff. Your extension request may also be denied. However, we will try to accommodate requests if they are reasonable and the new deadline does not extend past the date on which we will release solutions to the assignment on Ed.

Accommodations

We will provide appropriate accommodations to all students enrolled in Berkeley's [Disabled Students Program \(DSP\)](#). To ensure that you receive the appropriate accommodations, have your DSP specialist submit a letter confirming your status and accommodations.

If you're not enrolled in DSP, or are in the process of being onboarded by DSP, you may still be eligible for accommodations. We also aim to provide fair and appropriate accommodations to any students who, because of extenuating circumstances, may need them. Please reach out to data8@berkeley.edu in this case.

Privacy

All DSP and accommodations-related materials for this course are kept in a repository separate from the rest of the course materials that is visible only to the instructor. For any DSP and accommodations-related communications, please reach out to the instructor.

Do not cheat

It is important to keep in mind the limits of collaboration. As noted above, you and your peers are encouraged to discuss course content and approaches to problem solving. But you cannot share your code or written answers with other students. The two exceptions to this rule: you can share everything related to a project with your project partner (if you have one), and, if you are attending a lab session and have a lab partner, you can share everything related to that lab with your lab partner. Otherwise, you must write your answers in your own words and use your own

code, and you must not share your work.

In addition, **you must solve problems using the resources made available in the course and only the resources made available in the course!** This means that you are not permitted to turn in written or code answers that you have obtained from others, online sources *or from prior experience that does not include what we have taught in the course*. This also means that you are not permitted to submit written material or code created with any generative AI tools, including but not limited to ChatGPT. This also means that you may not submit assignments using code that we have not taught in the course. In addition, posting course content such as homeworks, projects, and exams on any 3rd party websites or submitting your own answers on outside sites/forums is considered academic misconduct.

This list of prohibited behaviors is not exhaustive and we reserve the right to punish other dishonest behaviors. We will report any academic dishonesty directly to the Center for Student Conduct, and you will immediately receive an F in the course. Please read Berkeley's [Code of Conduct](#) carefully.

Thanks for reading!

The main goal of the course is that you should learn and have a fantastic experience doing so. The other goal is, of course, to have fun. Welcome to Data 8!